ACCESS TO CARE

Opioid Overdose Risk & Treatment Availability Across U.S. Counties

CDC WONDER 2018–2024

SAMHSA Facility Data

U.S. Census ACS

USDA RUCC 2023

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1,887
counties analyzed

2018–2024
7-year study period

4
research questions

Research Questions

Four questions guiding the analysis

01 How do socioeconomic factors correlate with opioid overdose mortality?

Income · Poverty · Unemployment

02 Are treatment facilities distributed in proportion to overdose mortality rates?

Facility Rate vs Overdose Rate

03 Does treatment access correlate with lower overdose mortality?

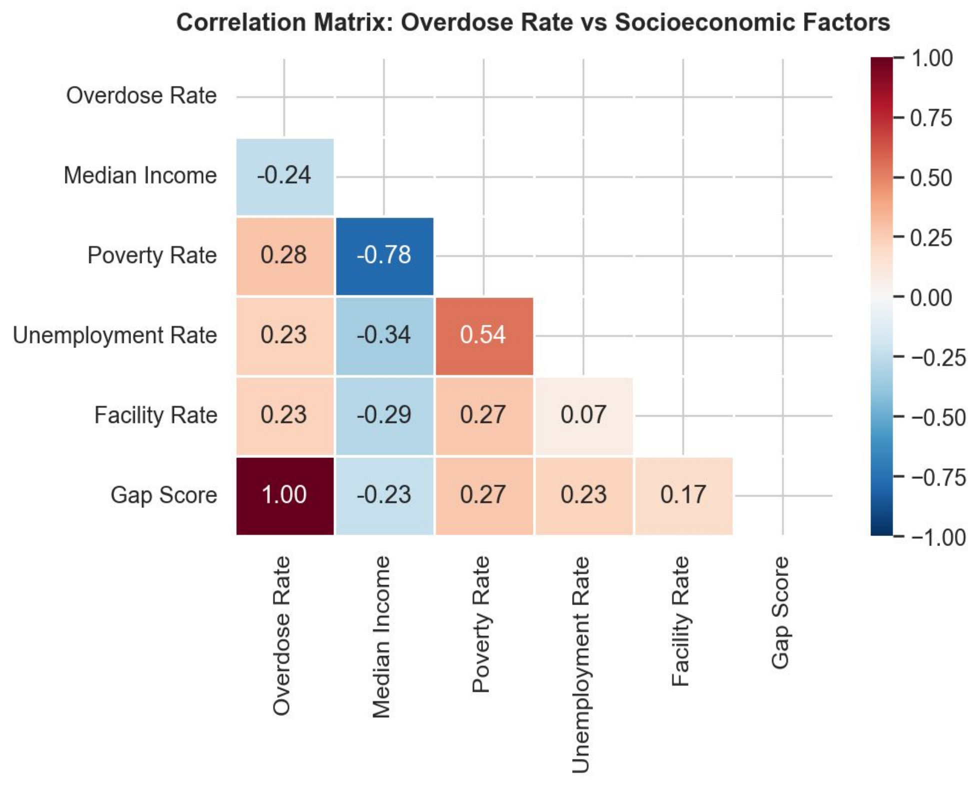
T-test · Multivariate regression · RUCC control

04 Which counties show the largest gaps between overdose risk and treatment availability?

Composite Risk Score · Gap Score

Q1: Socioeconomic Factors & Overdose Mortality

Pearson correlations — all $p < 0.001$ · $n = 1{,}887$ counties



$r = -0.24$

Income ↑ → Overdose ↓ (protective)

$r = +0.29$

Poverty ↑ → Overdose ↑

$r = +0.23$

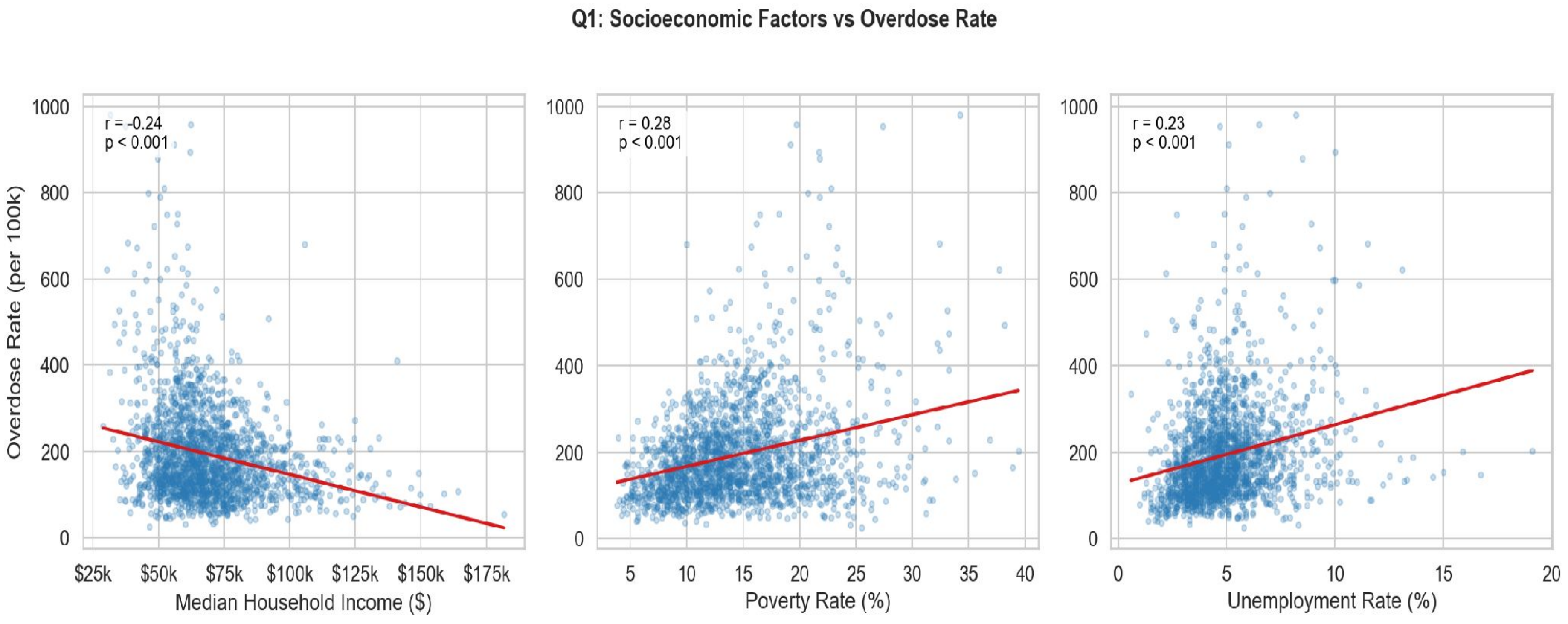
Unemployment ↑ → Overdose ↑

$p < 0.001$

All three findings statistically significant

Q1: Scatter Plots — Each Socioeconomic Factor vs Overdose Rate

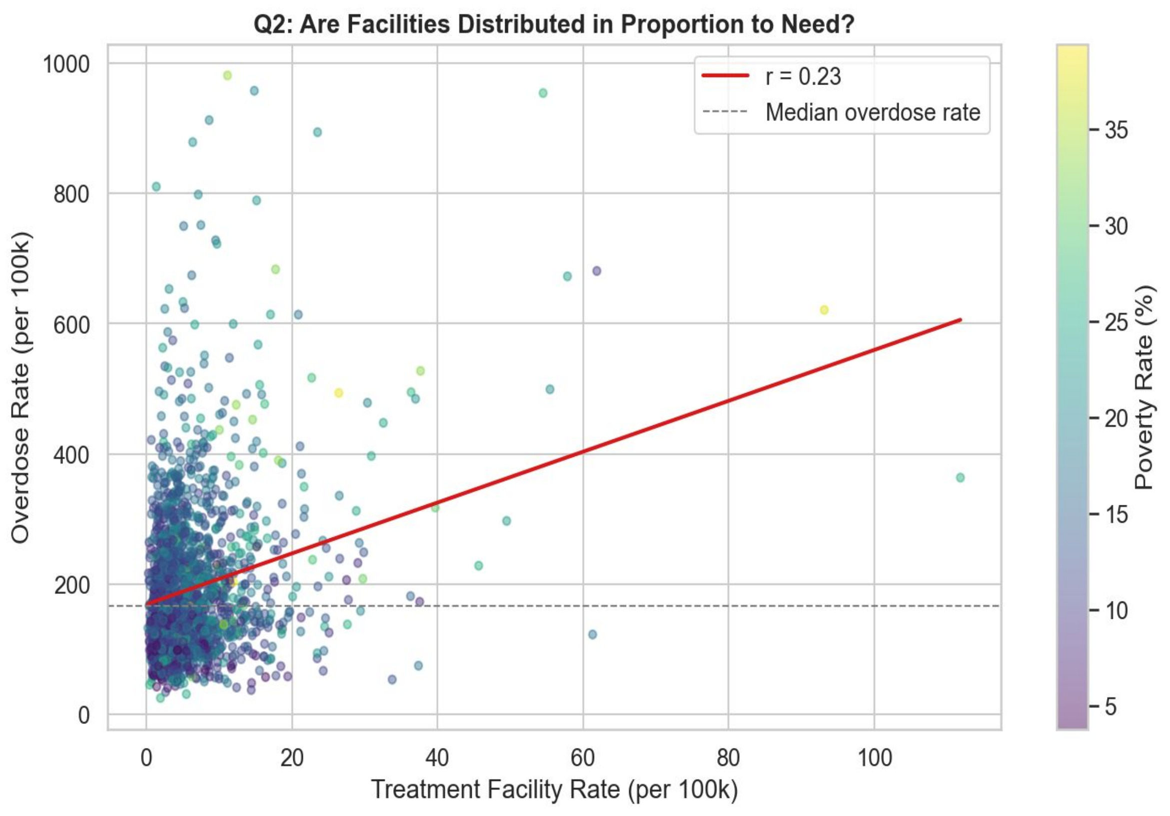
Each dot = one U.S. county · Red line = linear regression · All p < 0.001



Correlations are consistent in direction and significance but moderate in strength — socioeconomic factors explain roughly 8–12% of county-level variance individually, pointing to a multifactorial problem requiring broader intervention.

Q2: Are Facilities Distributed in Proportion to Need?

Facility rate vs overdose rate · color = poverty rate



Key Finding

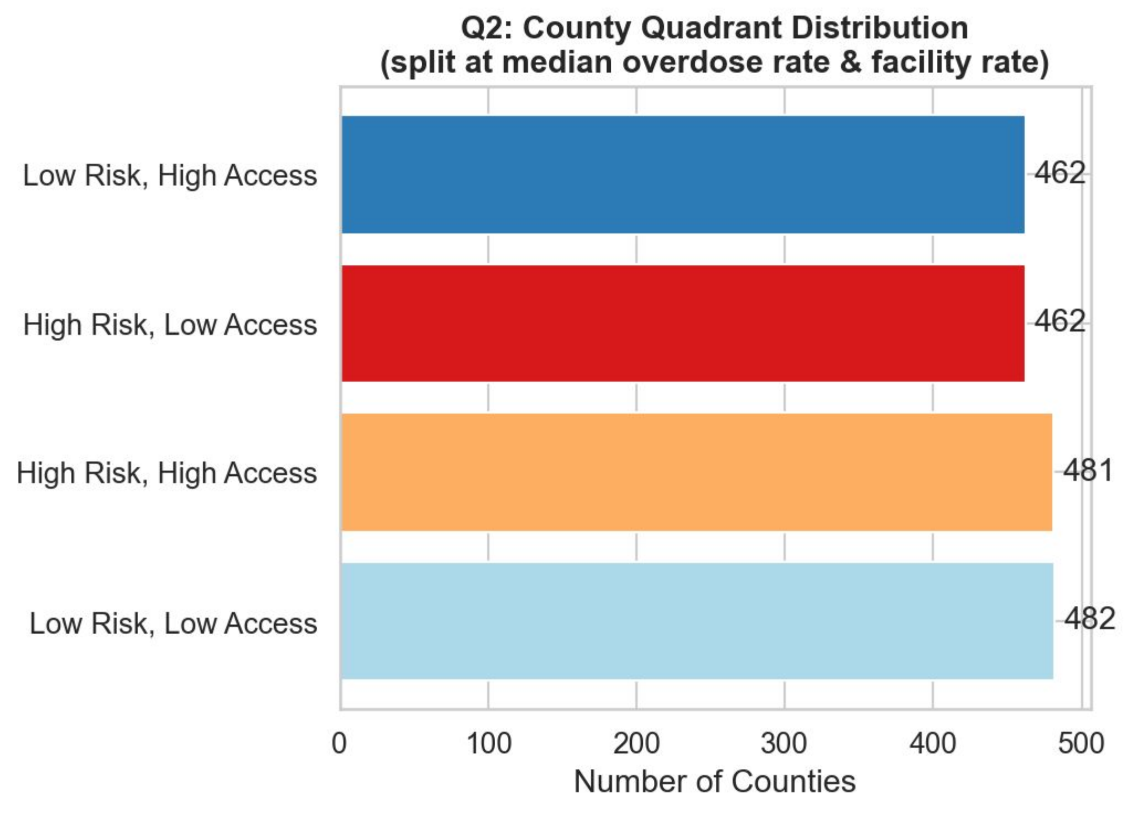
r = +0.23

If facilities tracked need, we'd expect a negative correlation. Instead r = +0.23 — facilities cluster where population is dense, not where overdose burden is highest. Higher-poverty counties (yellow) consistently appear in the high-overdose, low-facility zone.

Facilities follow population density, not overdose burden

Q2: County Quadrant Distribution

Split at median overdose rate & median facility rate



What this means

462 counties

are High Risk & Low Access — the primary target for policy intervention

Near-equal split

across all four quadrants shows facilities are not aligned with need

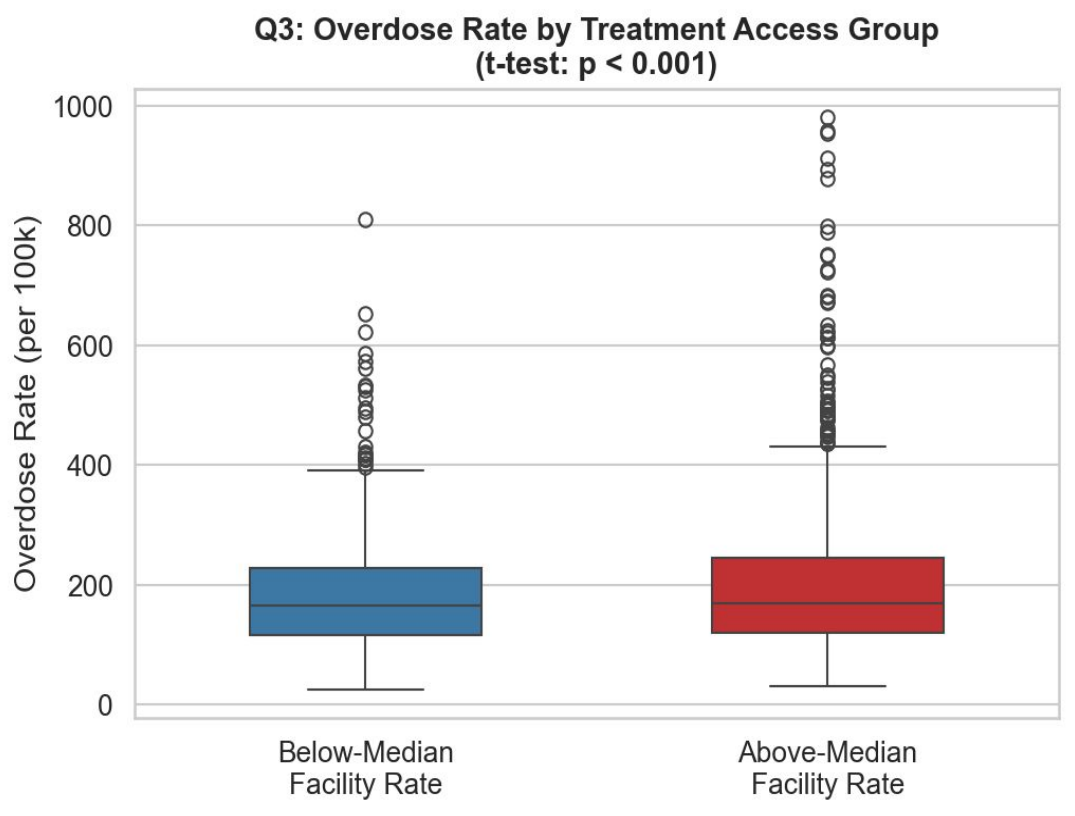
481 counties

have high risk AND high access — but mortality remains elevated

462 counties
High Risk · Low Access

Q3: Does Treatment Access Correlate with Lower Overdose Mortality?

Comparing counties above vs below median facility rate



Statistical Result

t-test:

$p < 0.001$ — statistically significant difference between groups

But note:

Boxplots show substantial overlap — the groups are not as different as the p-value alone implies

⚠️ **Confounders:**

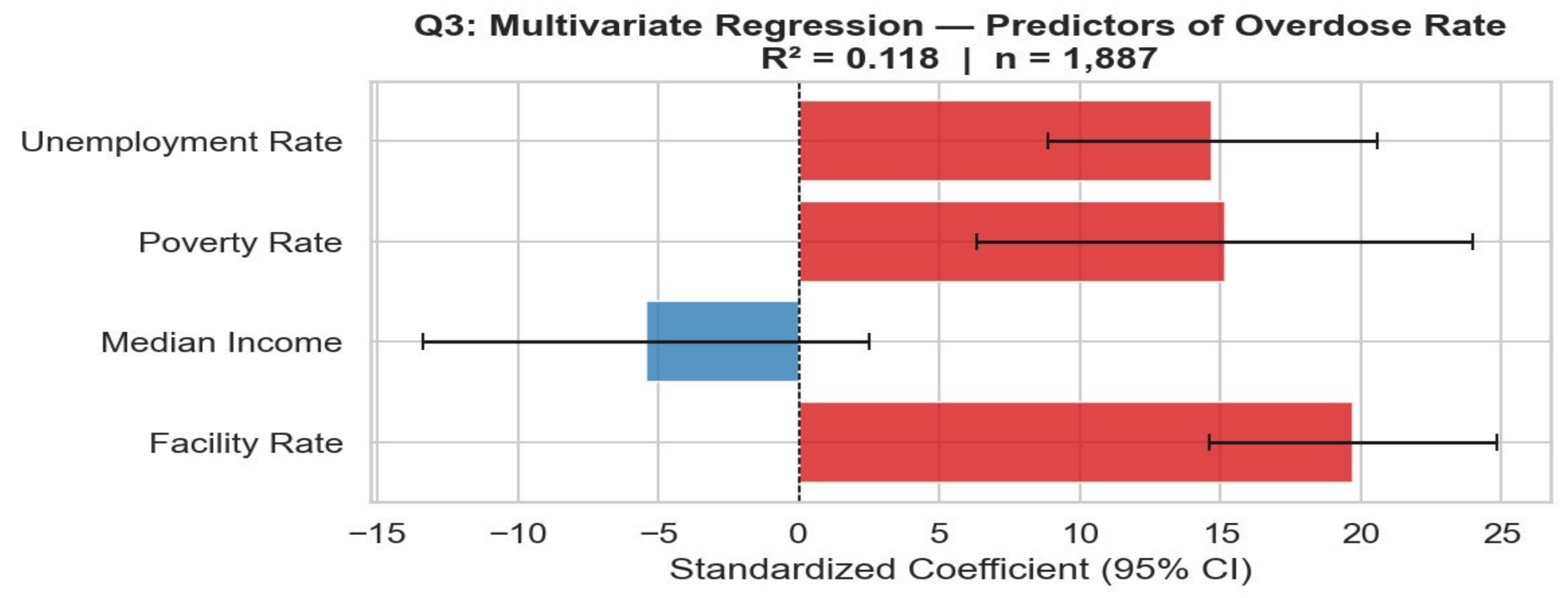
Higher-access counties tend to be more urban and wealthier — these factors also affect mortality

Next slides:

Multivariate regression and RUCC analysis isolate the true facility effect

Q3: Multivariate Regression — Original Model

Standardized coefficients · 95% CI · $R^2 = 0.118$ · n = 1,887

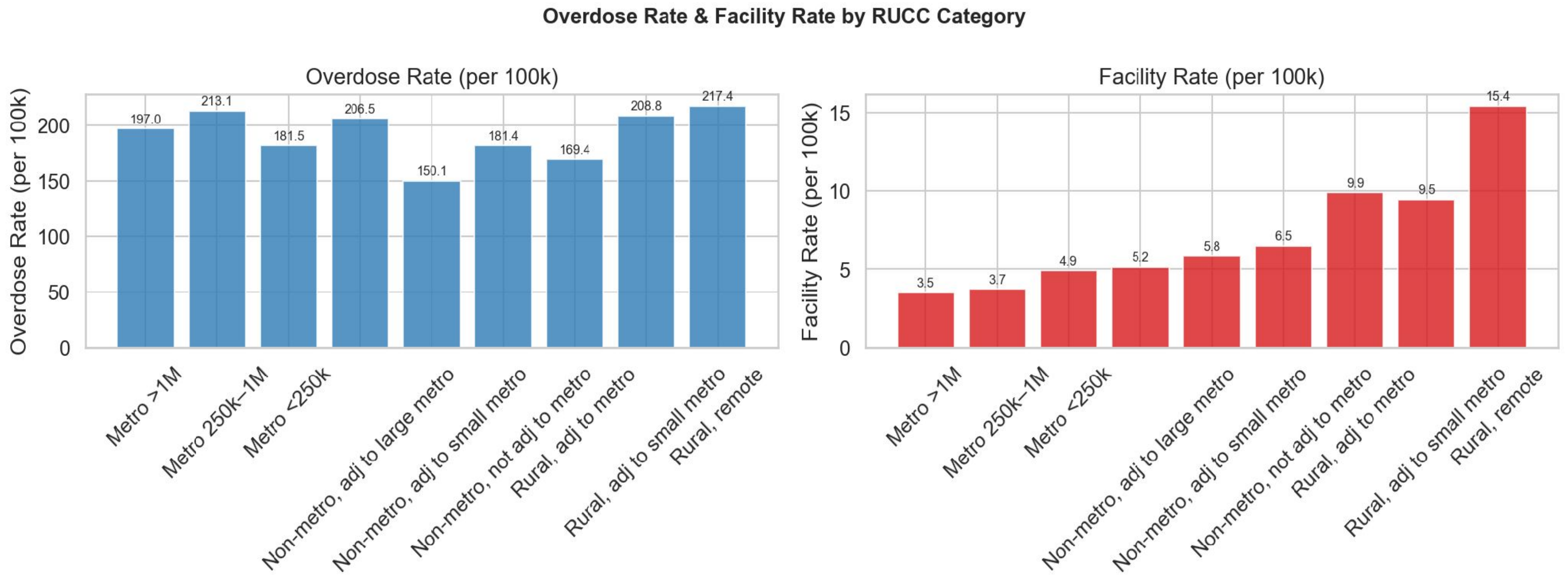


0.118	+19.7	+15.0	−5.1
R^2 — model explains 11.8% of variance	Facility Rate coef — counterintuitive positive	Unemployment Rate coef	Median Income coef — protective

⚠️ Facility Rate shows a positive coefficient — counterintuitive. Hypothesis: urban confounding. Counties with more facilities also tend to have more people and higher absolute overdose counts. Next slide tests this.

Q3 Extension: Overdose & Facility Rates by Urban–Rural Class. (RUCC)

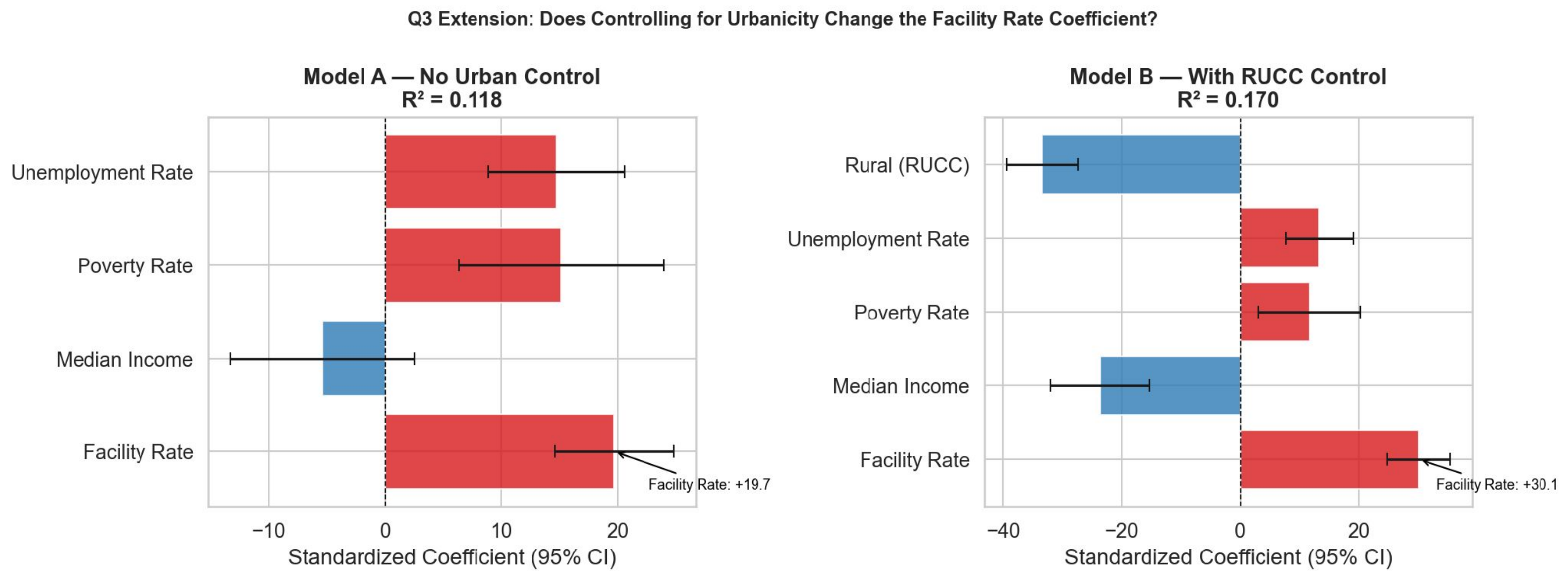
USDA Rural-Urban Continuum Codes · 1 = largest metro · 9 = most remote rural



⚠️ Overdose rates are relatively flat across RUCC categories (150–217 range) — rural counties are NOT lower-risk. But facility rates are 4× higher in remote rural counties than large metros on a per-100k basis, driven by small denominators. Rural counties face similar overdose burdens with far fewer absolute resources.

Q3 Extension: Controlling for Urbanicity — Model A vs Model B

Model A = original · Model B = adds RUCC · Both n = 1,887

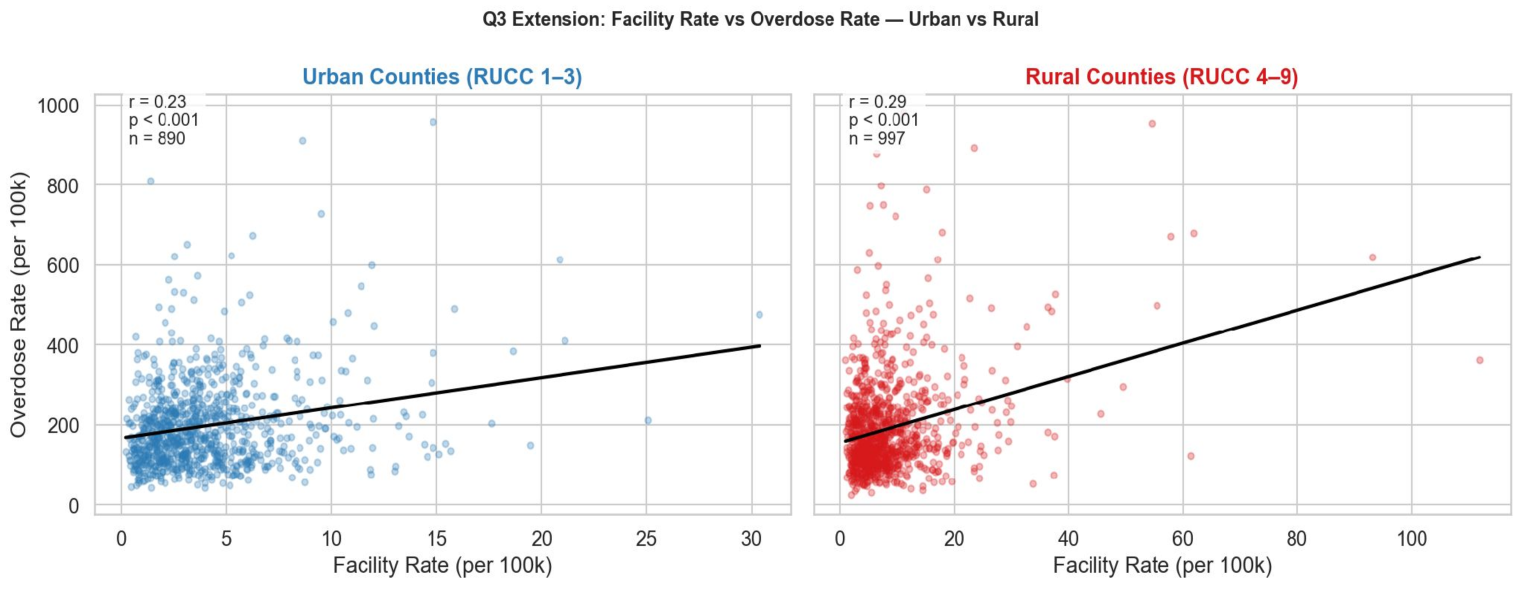


0.118	0.170	+19.7	+30.1	−33.4
R^2 — Model A	R^2 — Model B (+5.2%)	Facility Rate — Model A	Facility Rate — Model B	RUCC coef — more rural = higher OD rate

RUCC coefficient of −33.4 ($p < 0.001$): rural counties face significantly higher overdose mortality independent of income, poverty, and unemployment. R^2 improved from 0.118 ⟶ 0.170. The facility rate coefficient did not flip — facilities ARE placed in high-overdose counties within each urbanicity tier, but are not reducing mortality. This is a capacity and accessibility problem, not a placement problem alone.

Q3 Extension: Facility Rate vs Overdose Rate — Urban vs Rural

Positive correlation holds within BOTH groups — ruling out simple urban confounding



r = 0.23
Urban counties (RUCC 1–3) · n = 890

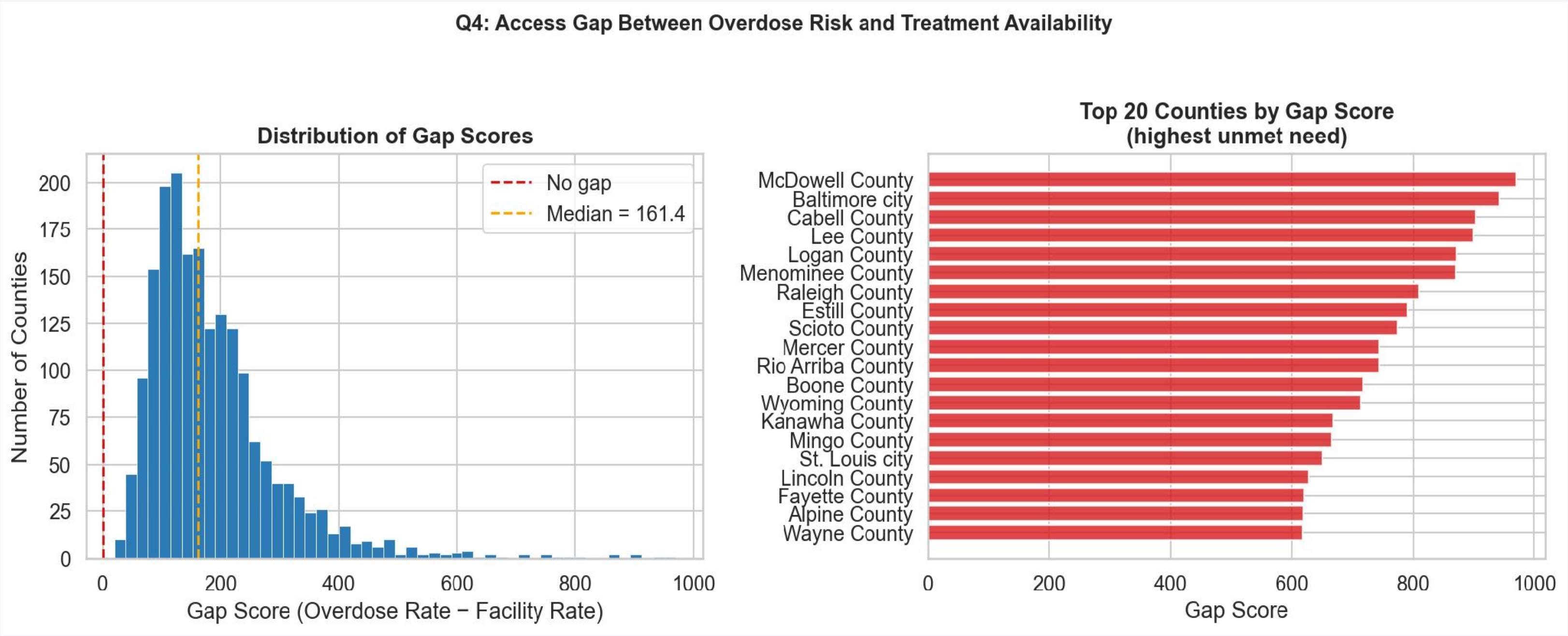
r = 0.29
Rural counties (RUCC 4–9) · n = 997

p < 0.001
Both groups statistically significant

The relationship is NOT an artifact of mixing urban and rural counties — it holds within both groups independently. Conclusion: facilities are placed reactively (where need is highest) but supply is not keeping pace with demand in either setting.

Q4: Access Gap Between Overdose Risk & Treatment Availability

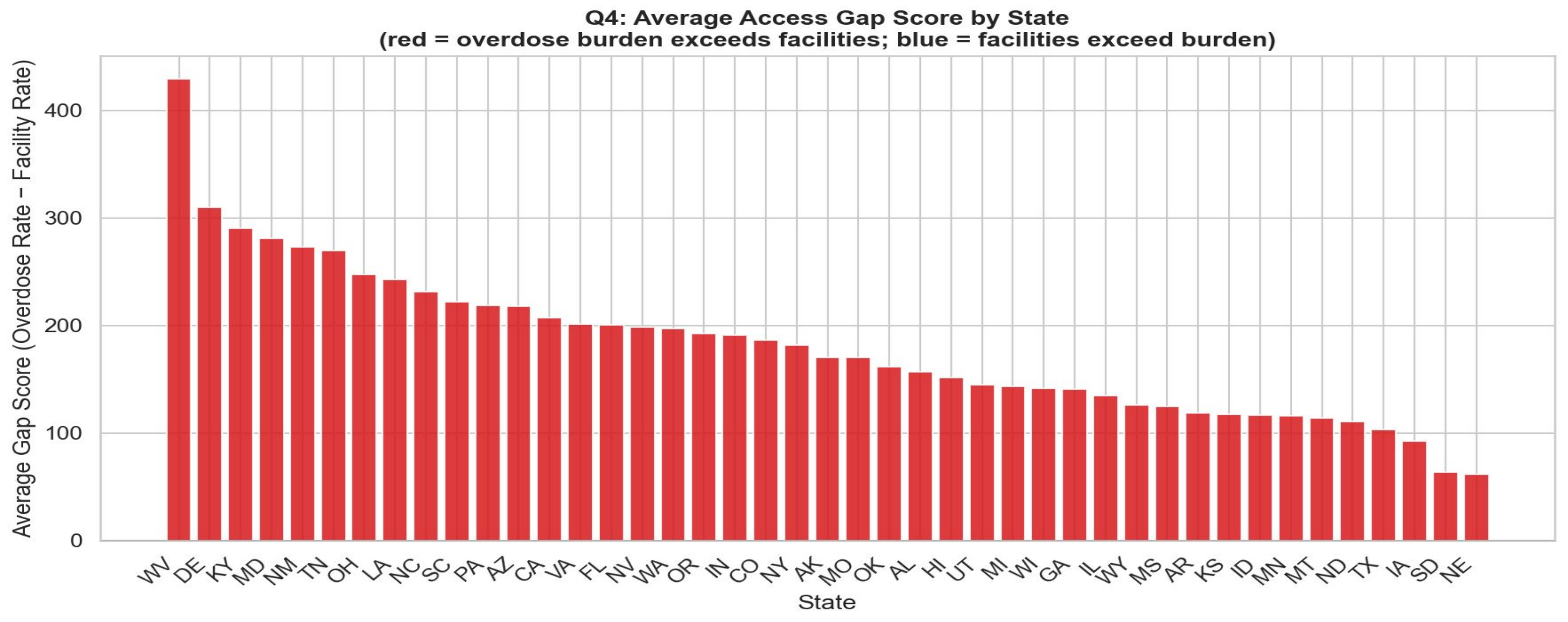
Gap Score = Overdose Rate − Facility Rate · Median gap = 161.4



Nearly every county has a positive gap score — overdose burden consistently exceeds treatment supply nationwide. The distribution is right-skewed: most counties cluster around 100–200 but a long tail of severely underserved counties extends toward 1,000. McDowell County, WV leads at nearly 1,000.

Q4: Average Access Gap Score by State

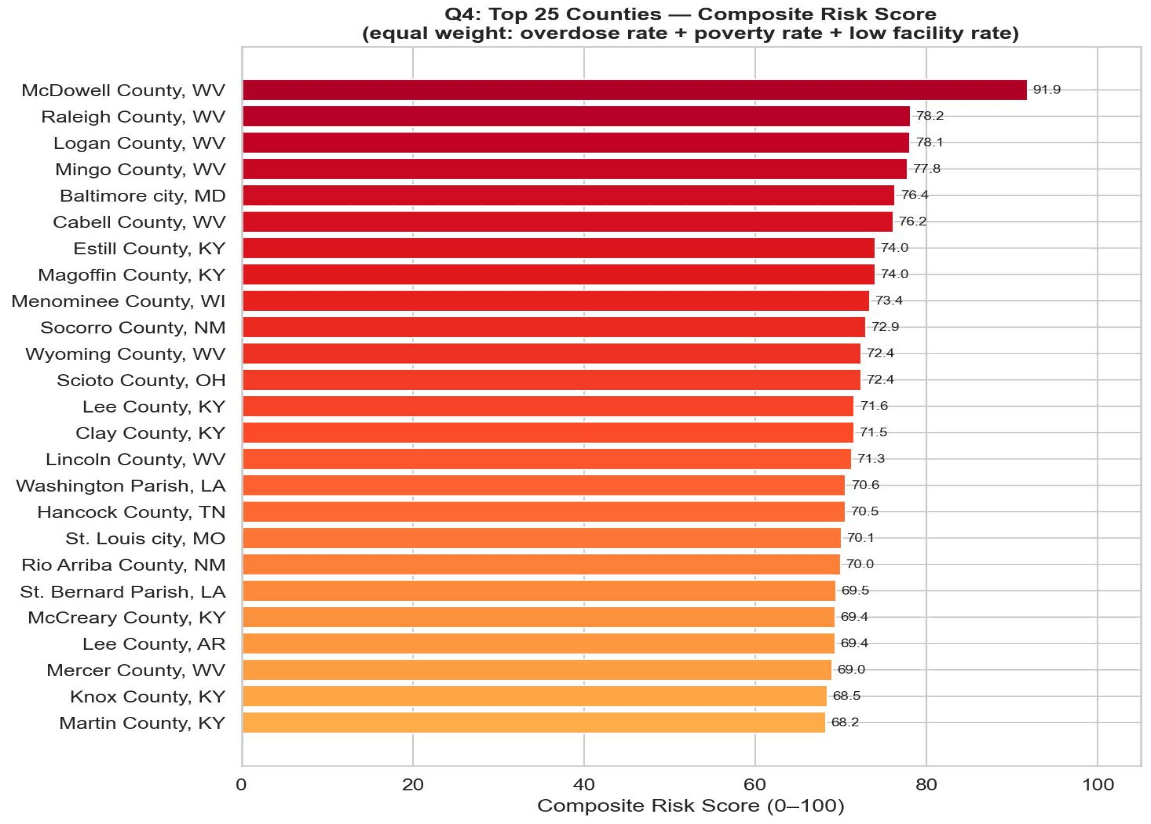
All states show positive gaps — overdose burden exceeds treatment supply everywhere



WV ~430	DE ~310	KY ~290	0 states
Highest avg gap score	2nd highest	3rd highest	Have negative gap (no state has excess capacity)

Q4: Top 25 Most At-Risk Counties — Composite Risk Score

Equal weight: normalized overdose rate + poverty rate + low facility rate · scaled 0–100



Patterns in the Top 25

🥇 **McDowell County, WV**

Score 91.9/100 — highest composite risk in the nation

📍 **West Virginia**

7 of top 25 counties — dominates due to high overdose rate, high poverty, and low facilities simultaneously

📍 **Kentucky**

7 counties (Estill, Magoffin, Lee, Clay, McCreary, Knox, Martin) — concentrated in Appalachia

🏙️ **Baltimore City, MD**

Only major urban area in top 25 — shows this is not exclusively a rural problem

🗺️ **Pattern**

Appalachian + rural Southern counties dominate — all three risk factors align simultaneously

Interactive Dashboard

Built with R Shiny · Leaflet · Plotly · DT · USDA RUCC Integration

🗺️ Choropleth Map

9 selectable metrics including RUCC urban-rural classification. Click any county for full stats popup including gap score and RUCC category.

⚠️ At-Risk Table

Composite risk score table with user-controlled slider (5–100 counties). Filters apply — isolate the most at-risk rural counties in a specific state.

🔍 Dual Filters

Filter by state AND urban-rural type (Metro / Non-metro / Rural) simultaneously to focus on any geography.

📊 County Rankings

Top 20 bar chart by any metric. Value boxes update dynamically including "Rural Counties: High Risk & Low Access" count.

Key Takeaways

01 Poverty & unemployment drive overdose risk

r = +0.29 poverty, r = +0.23 unemployment, r = −0.24 income — all $p < 0.001$. Socioeconomic vulnerability is a consistent, significant predictor.

03 Rural counties face compounded disadvantage

After controlling for income, poverty, and unemployment, rurality independently predicts higher overdose mortality (RUCC coef = −33.4, $p < 0.001$, R^2 0.118 → 0.170).

02 Facilities are misallocated — 462 counties left behind

462 counties are High Risk & Low Access. Facility placement follows population density, not overdose burden. The distribution is nearly random relative to need.

04 Appalachian counties face the highest combined risk

McDowell County, WV scores 91.9/100 on the composite risk index. WV and KY together account for 14 of the top 25 most at-risk counties nationwide.

Limitations & Future Directions

Honest assessment of what this analysis can and cannot conclude

⚠️ Limitations

CDC WONDER suppression

Counties with <10 deaths have missing data — rural low-mortality areas are underrepresented, potentially understating rural risk

Correlational only

No causal claims can be made. Unmeasured confounders (drug supply, state policy, cultural factors) likely explain remaining variance

Facility presence ≠ access

SAMHSA data captures facility existence, not capacity, cost, wait times, or whether patients can actually receive care

ACS time alignment

5-year ACS estimates may not perfectly align with the 2018–2024 mortality observation period

🔭 Future Directions

Add drug supply data

Incorporate DEA fentanyl seizure data by county to separate supply-side from demand-side drivers

State policy variables

Include naloxone access laws, Medicaid expansion status, and opioid prescribing regulations as covariates

Longitudinal analysis

Track how counties move across risk quadrants over time as policy and facility access change

Capacity-adjusted access

Replace facility count with capacity-weighted or utilization-based measures for a truer access metric